

Mohd Uwaish

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TECHNICAL SKILLS

Languages : Python, C, JavaScript, SQL, PL/SQL

ML & DL : Scikit-Learn, PyTorch, TensorFlow, XGBoost, Neural Networks, Transformers

NLP & LLMs : NLTK, SpaCy, Word2Vec, HuggingFace, Prompt Engineering, Fine-Tuning

RAG & LLM Frameworks : LangChain, LangGraph, LlamaIndex, RAGAS, Retrieval-Augmented Generation

Data & Visualisation : NumPy, Pandas, Plotly, Matplotlib, Seaborn, Tableau

Databases & Vector Stores : MySQL, MongoDB, SQL Server, ChromaDB, Pinecone

Web Frameworks : Django, Flask, Streamlit, FastAPI, React, Next.js

DevOps & Deployment : Git, GitLab, GitHub, Docker, AWS, GCP, CI/CD

PROFESSIONAL EXPERIENCE

Fraunhofer IEG

Sep. 2025 – Present

Research Assistant – Deep Learning and NLP

Bochum, Germany

- Conducted literature research on integrating news data into time series models for **electricity price and demand forecasting**, identifying gaps in cross-modal feature fusion approaches
- Developed **web scraping pipelines** to collect news articles from multiple Turkish news sources with automated neural machine translation to English
- Extracted NLP features including **sentiment scores via VADER and BERT**, named entities using fine-tuned BERT, and latent topics using LDA and BERTopic
- Designed and executed **15+ experiments** integrating extracted news features with historical data across XGBoost, LSTM, and Transformer architectures
- Concluded that news-derived features negatively impacted model performance, attributing the result to the high volatility of electricity data outpacing the temporal resolution of news signals

Georg-August-Universitaet Goettingen

Feb. 2024 – Present

Student Assistant – Department of Economics

Goettingen, Germany

- Developing user-friendly economics experiments using frontend technologies and the **Django-based oTree framework**
- Developed an **oTree Proxy server** connecting the experiment system with Prolific to launch scalable experiments
- Implementing **chat and video communication features** to enhance participant interaction, experimental realism, and data collection
- Conducting data analysis for experiments, employing statistical methods and visualization techniques to derive insights

Niedersaechsische Staats- und Universitaetsbibliothek Goettingen

Sep. 2024 – Aug. 2025

Student Assistant – Software Developer

Goettingen, Germany

- Developed the **DFG-funded digital library web portal** for Bibliothek der Neologie using Next.js with hybrid critical edition features and bilingual (German/English) content management
- Performed detailed pull request reviews to ensure code quality, maintainability, and adherence to project standards, **reducing integration bugs**
- Authored and maintained structured technical documentation significantly **improving team productivity** and knowledge transfer
- Utilized Git/GitLab and **CI/CD pipelines** to streamline development and ensure code quality

Tata Consultancy Services

Sep. 2021 – Apr. 2023

Data Analyst and Software Engineer

New Delhi, India

- Developed full-stack analytics solutions for the PENALTY module of ITBA using Java, PL/SQL, Oracle DB, Tableau, and React.js, resulting in a **23% increase in user engagement** and **17% reduction in processing time**
- Designed and optimized ETL pipelines and PL/SQL workflows achieving a **62% improvement in report generation time** and 18% faster database response through advanced query tuning
- Collaborated with cross-functional teams, contributed to **9 successful change requests**, and mentored peers to **reduce bug rates by 30%**

PROJECTS

FairSample: Fair Sampling Library for Imbalanced Datasets | *Open-Source ML Python Package* | [GitHub](#)

- Designed and published an **open-source Python package** implementing 14+ resampling techniques (RFCL, EHSO, ODBOT, OSM, SVD-DWSMOTE, KMeans-based) with a unified scikit-learn-compatible API for handling class imbalance
- Implemented **40+ dataset complexity measures** across feature overlap (F1, F2, F3), instance overlap (N3, N4, kDN), and structural categories (T1, DBC, LSC) to diagnose *why* a dataset is difficult before resampling
- Built a pre/post resampling evaluation framework enabling quantitative comparison of overlap reduction and class separability improvements across techniques
- Packaged and published to **PyPI** with full documentation; installable via `pip install fairsample` for reproducible research workflows

GeoRAG: Text Information Extraction from Geochemical Papers | *Math Info Retrieval Seminar* | [GitHub](#)

- Designed a novel **Hybrid RAG system** combining vector-based semantic retrieval with graph-based structured knowledge extraction
- Developed a dynamic query classification mechanism to optimize retrieval strategies based on query intent, improving contextual accuracy by **23% over baseline**
- Engineered a **Neo4j knowledge graph pipeline** capturing geochemical relationships between elements, isotopes, and minerals for enhanced reasoning
- Implemented comprehensive evaluation framework with **11 metrics** assessing retrieval quality and response generation; demonstrated significant improvements in factual correctness, semantic similarity, and noise reduction

Google Chrome Extension – Real-Time Sensitive Image Detection | *Web Browser Extension* | [GitHub](#)

- Built a cross-platform Chrome extension (ReactJS + Chrome APIs) for real-time sensitive image detection on Facebook, Instagram, and X (Twitter)
- Developed a Flask-based backend hosting a **fine-tuned BEiT model (F1 score: 77.48%)** trained on 12K+ labeled images from the SensitivAlert Pic dataset
- Integrated frontend-backend via Base64 image encoding and REST API with asynchronous communication using Chrome's messaging system
- Designed interactive UI with prediction results, override dialog, and feedback form; user feedback stored in MongoDB for model improvement
- Implemented **auto-retraining pipeline** triggered at 50+ flagged samples using active learning and augmentation (cropping, color jitter, rotation)

Universitaet Kompass: AI-Powered University Program Recommendation | *AI/NLP Web App* | [GitHub](#)

- Developed an AI web application using Python, Streamlit, FAISS, BeautifulSoup, and MongoDB to **recommend best-matching German university programs** based on user-uploaded resumes
- Built a web scraping pipeline to extract and normalize program data, contacts, and deadlines from DAAD and official university portals
- Integrated **OpenAI GPT-4 and NLP techniques** to extract structured information from PDF resumes for personalized matching
- Implemented semantic search and ranking using **FAISS** for efficient program retrieval and scoring
- Enabled users to view personalized program suggestions with detailed info, contact data, and application deadlines via a dynamic Streamlit interface

Meeting Summarization Testbench | *NLP Evaluation and Analysis* | [GitHub](#)

- Developed a comprehensive testbench for evaluating meeting summarization models on transcripts with a dataset augmentation pipeline using **17 augmentation techniques**
- Designed a full-stack system using Django REST Framework and HTML/CSS/JS allowing users to upload Hugging Face models for detailed evaluation
- Conducted metric evaluation including **ROUGE, BLEU, BERTscore**, and linguistic analysis (type-token ratio, Coleman Liau index, coverage) to assess summary quality
- Utilized visualization techniques (histograms and charts) to present evaluation results; implemented with Hugging Face, Pandas, and Plotly for scalability

EDUCATION

Georg-August-Universitaet Goettingen

M.Sc. in Applied Computer Science, Specialization: Data Science

Apr. 2023 – Present
Goettingen, Germany

- **Grade:** 2.0 (current)

Guru Gobind Singh Indraprastha University

B.Tech in Computer Science and Engineering

May 2017 – May 2021
New Delhi, India

- **CGPA:** 8.01/10
- **Thesis:** Eth-Ocracy – Ethereum Blockchain-based Voting System (Full-stack, PostgreSQL, Celery, RabbitMQ, Selenium)